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Connector and connector device

Abstract

A first connector according to one aspect of the present disclosure capable of electrically connecting a first connection terminal of a first in-vehicle device and a second electrical contact portion of a second in-vehicle device is provided with a first terminal fitting and a first connector housing for accommodating the first terminal fitting. The first terminal fitting includes a first device-side connecting portion connectable to the first connection terminal and a first electrical contact portion connectable to the second electrical contact portion and arranged on a side opposite to the first device-side connecting portion in a length direction of the first terminal fitting. The first electrical contact portion and the first device-side connecting portion are constituted by a single component. The first electrical contact portion is relatively displaceable with respect to the first device-side connecting portion.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is a national phase of PCT application No. PCT/JP2021/041509, filed on 11 Nov. 2021, which claims priority from Japanese patent application Nos. 2020-188801 and 2021-137461, filed on 12 Nov. 2020 and 25 Aug. 2021, respectively, all of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to a connector and a connector device.

BACKGROUND

(3) Conventionally, a structure has been adopted which, to electrically connect in-vehicle devices such as a motor and an inverter, connects a connector for external connection provided in one device and a connector for external connection provided in the other device via a wiring harness. In recent years, a connector device has been proposed which directly connects connectors for external connection of both devices to cope with requests such as a reduction in the number of components and space saving.

(4) For example, in a connector device disclosed in Patent Document 1, a connector provided in one device includes a terminal fitting, a connector housing and a supporting member, and the terminal fitting includes an electrical contact portion to be connected to a mating connector, a device-side connecting portion to be connected to an in-vehicle device and an electrical coupling portion for coupling those portions. The supporting member supports the electrical contact portion displaceably in the connector housing. Thus, even if the electrical contact portion of the connector is shifted in position from that of the mating connector, the electrical contact portion supported by the supporting member is displaced and deflected in the connector housing, whereby a position shift is absorbed and the electrical contact portions of the both connectors are electrically connected.

PRIOR ART DOCUMENT

Patent Document

(5) Patent Document 1: JP 2010-225488 A

SUMMARY OF THE INVENTION

Problem to be Solved

(6) In the connector device of Patent Document 1, the terminal fitting in the connector of the one device is provided by joining a plurality of members including the electrical coupling portion constituted by a braided wire or the like. Further, a structure for holding the supporting member supporting the electrical contact portion displaceably in the connector housing is also necessary. As a result, problems of a large number of components, a complicated structure and an unavoidable cost increase have been inherent.

(7) Accordingly, a connector and a connector device of novel structures are disclosed which enable a reduction in the number of components, the simplification of a structure and a cost reduction.

Means to Solve the Problem

(8) The present disclosure is directed to a connector capable of electrically connecting a first terminal portion of a first in-vehicle device and a mating electrical contact portion of a second in-vehicle device, the connector being provided with a first terminal fitting, and a first connector housing for accommodating the first terminal fitting, the first terminal fitting including a first device-side connecting portion connectable to the first terminal portion and a first electrical contact portion connectable to the mating electrical contact portion and arranged on a side opposite to the first device-side connecting portion in a length direction of the first terminal fitting, the first electrical contact portion and the first device-side connecting portion being constituted by a single

component, and the first electrical contact portion being relatively displaceable with respect to the first device-side connecting portion.

(9) The present disclosure is also directed to a connector device for electrically connecting a first terminal portion of a first in-vehicle device and a second terminal portion of a second in-vehicle device, the connector device being provided with a first connector mountable into the first in-vehicle device and a second connector mountable into the second in-vehicle device, the first connector including a first terminal fitting and a first connector housing for accommodating the first terminal fitting, the second connector including a second terminal fitting, the second terminal fitting including a second device-side connecting portion connectable to the second terminal portion and a second electrical contact portion to be connected to the first terminal fitting and arranged on a side opposite to the second device-side connecting portion, the first terminal fitting including a first device-side connecting portion connectable to the first terminal portion and a first electrical contact portion to be connected to the second electrical contact portion and arranged on a side opposite to the first device-side connecting portion in a length direction of the first terminal fitting, the first electrical contact portion and the first device-side connecting portion being constituted by a single component, and the first electrical contact portion being relatively displaceable with respect to the first device-side connecting portion.

Effect of the Invention

(10) According to the connector and the connector device of the present disclosure, it is possible to achieve a reduction in the number of components, the simplification of a structure and a cost reduction.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is an overall perspective view showing a connector device according to one embodiment.
- (2) FIG. 2 is a longitudinal section of the connector device shown in FIG. 1.
- (3) FIG. 3 is an exploded perspective view of the connector device shown in FIG. 1.
- (4) FIG. 4 is an exploded perspective view of the connector device shown in FIG. 3 showing a connected state of a first terminal and a second terminal.
- (5) FIG. 5 is a partial section of the connector device shown in FIG. 1.
- (6) FIG. 6 is a partial section of the connector device shown in FIG. 1.
- (7) FIG. 7 is a partial section of the connector device shown in FIG. 1.
- (8) FIG. 8 is a partial section of the connector device shown in FIG. 1.
- (9) FIG. 9 is a partial section of the connector device shown in FIG. 1.
- (10) FIG. 10 is a partial section of the connector device shown in FIG. 1.
- (11) FIG. 11 is a partial section of a connector device according to another example.
- (12) FIG. 12 is a partial exploded perspective view of the connector device according to the other example.
- (13) FIG. 13 is a longitudinal section of a connector device according to another example.
- (14) FIG. 14 is a longitudinal section of a connector device according to another example.

DETAILED DESCRIPTION TO EXECUTE THE INVENTION

Description of Embodiments of Present Disclosure

(15) First, embodiments of the present disclosure are listed and described.

(16) (1) The connector of the present disclosure is a connector capable of electrically connecting a first terminal portion of a first in-vehicle device and a mating electrical contact portion of a second in-vehicle device, and provided with a first terminal fitting and a first connector housing for accommodating the first terminal fitting, the first terminal fitting including a first device-side

connecting portion connectable to the first terminal portion and a first electrical contact portion connectable to the mating electrical contact portion and arranged on a side opposite to the first device-side connecting portion in a length direction of the first terminal fitting, the first electrical contact portion and the first device-side connecting portion being constituted by a single component, and the first electrical contact portion being relatively displaceable with respect to the first device-side connecting portion.

(17) According to the connector of the present disclosure, a position shift from the mating electrical contact portion can be absorbed, using a displacement of the first electrical contact portion with respect to the first device-side connecting portion. The first electrical contact portion and the first device-side connecting portion are constituted by the single component. Thus, a reduction in the number of components and the simplification of a structure can be advantageously achieved as compared to a conventional structure in which a terminal fitting needs to be constituted by joining a plurality of members including a braided wire and a supporting member for supporting an electrical contact portion and a mechanism for displaceably holding the supporting member in a connector housing are necessary.

(18) (2) Preferably, the first terminal fitting includes a first stretchable portion between the first device-side connecting portion and the first electrical contact portion, and the first electrical contact portion is relatively displaceable with respect to the first device-side connecting portion by resilient deformation of the first stretchable portion. Since the first terminal fitting includes the resiliently deformable first stretchable portion in an intermediate part in the length direction of the first terminal fitting, position shifts of the first electrical contact portion and the mating electrical contact portion can be more advantageously absorbed.

(19) (3) Preferably, the first stretchable portion is formed by bending the first terminal fitting. The first stretchable portion can be easily provided by bending the first terminal fitting and configured without increasing the number of components.

(20) (4) Preferably, a resilient member is provided which is accommodated into the first connector housing, and the resilient member is capable of holding the first electrical contact portion in contact with the mating electrical contact portion and displaceable according to a displacement of the first electrical contact portion. By including the resilient member capable of holding the first electrical contact portion in contact with the mating electrical contact portion, conduction stability between the both electrical contact portions can be more reliably maintained. Further, since the resilient member is accommodated in the first connector housing in a state displaceable according to the displacement of the first electrical contact portion, it is advantageously prevented by the resilient member that a position shift absorbing function is hindered by the displacement of the first electrical contact portion.

(21) (5) Preferably, the resilient member includes a pair of flat plate portions arranged to face each other with a gap therebetween and a coupling plate portion coupling the pair of flat plate portions, and the resilient member is capable of sandwiching the first electrical contact portion and the mating electrical contact portion in a contact state in mutually contacting directions by the pair of flat plate portions. In this way, the structure of the first connector can be further simplified.

(22) (6) Preferably, each of the pair of flat plate portions includes a sandwiching protrusion projecting inward in a facing direction, the sandwiching protrusions are capable of sandwiching the first electrical contact portion and the mating electrical contact portion in the contact state in the mutually contacting directions, and the sandwiching protrusions have an arcuate cross-section. Since the sandwiching protrusions for sandwiching the first electrical contact portion and the mating electrical contact portion have an arcuate cross-section, the first electrical contact portion or the mating electrical contact portion and the flat plate portion are allowed to be inclined.

(23) (7) Preferably, the first electrical contact portion includes a dome-shaped projecting contact point portion, and the contact point portion is capable of contacting the mating electrical contact portion. Since the contact point portion provided at the first electrical contact portion has a dome

shape, a contact state between the both electrical contact portions can be stably maintained even if the mating electrical contact portion to be overlapped on the first electrical contact portion is twisted and displaced such as by being rolled.

(24) (8) A connector device of the present disclosure is a connector device for electrically connecting a first terminal portion of a first in-vehicle device and a second terminal portion of a second in-vehicle device, and provided a first connector mountable into the first in-vehicle device and a second connector mountable into the second in-vehicle device, the first connector including a first terminal fitting and a first connector housing for accommodating the first terminal fitting, the second connector including a second terminal fitting, the second terminal fitting including a second device-side connecting portion connectable to the second terminal portion and a second electrical contact portion to be connected to the first terminal fitting and arranged on a side opposite to the second device-side connecting portion, the first terminal fitting including a first device-side connecting portion connectable to the first terminal portion and a first electrical contact portion to be connected to the second electrical contact portion and arranged on a side opposite to the first device-side connecting portion in a length direction of the first terminal fitting, the first electrical contact portion and the first device-side connecting portion being constituted by a single component, and the first electrical contact portion being relatively displaceable with respect to the first device-side connecting portion.

(25) According to the connector device of the present disclosure, a position shift from the second electrical contact portion can be absorbed, using a displacement of the first electrical contact portion with respect to the first device-side connecting portion. The first electrical contact portion and the first device-side connecting portion are constituted by the single component. Thus, a reduction in the number of components and the simplification of a structure can be advantageously achieved as compared to a conventional structure in which a terminal fitting needs to be constituted by joining a plurality of members including a braided wire and a supporting member for supporting an electrical contact portion and a mechanism for displaceably holding the supporting member in a connector housing are necessary.

(26) (9) Preferably, the second electrical contact portion and the second device-side connecting portion are constituted by a single component, and the second electrical contact portion is relatively displaceable with respect to the second device-side connecting portion. A position shift from the first electrical contact portion can be absorbed, using a displacement of the second electrical contact portion with respect to the second device-side connecting portion. The second electrical contact portion and the second device-side connecting portion are constituted by the single component. Therefore, a reduction in the number of components and the simplification of the structure can be more advantageously achieved.

(27) (10) Preferably, the second electrical contact portion projects to outside of the second in-vehicle device with the second connector mounted in the second in-vehicle device. Since the second electrical contact portion projects to the outside of the second in-vehicle device in a state relatively displaceable with respect to the second device-side connecting portion, position shifts between the respective electrical contact portions of the first and second connectors can be more advantageously absorbed and connection between the both electrical contact portions can be stably realized, using displacements of the both first and second electrical contact portions at the time of connecting the first and second connectors.

(28) (11) Preferably, the first terminal fitting includes a first stretchable portion between the first device-side connecting portion and the first electrical contact portion, the first electrical contact portion is relatively displaceable with respect to the first device-side connecting portion by resilient deformation of the first stretchable portion, and the first stretchable portion is formed by bending the first terminal fitting. Since the first terminal fitting includes the resiliently deformable first stretchable portion in an intermediate part in the length direction of the first terminal fitting, position shifts of the first and second electrical contact portions can be more advantageously

absorbed. The first stretchable portion can be easily provided by bending the first terminal fitting and configured without increasing the number of components.

(29) (12) Preferably, the first terminal fitting is thinner than the second terminal fitting. Since the first terminal fitting is thinner than the second terminal fitting, the first stretchable portion is easily resiliently deformed.

(30) (13) Preferably, a holding force between the first and second electrical contact portions is set to be larger than a force necessary for the resilient deformation of the first stretchable portion. For example, when the first and second in-vehicle devices are relatively displaced due to vibration or the like during the travel of a vehicle, the first stretchable portion is resiliently deformed earlier than relative displacements of the first and second electrical contact portions in the length direction of the first terminal fitting. Thus, the wear of the first and second electrical contact portions is suppressed.

(31) (14) Preferably, the holding force between the first and second electrical contact portions includes a friction force between members contacting in a direction intersecting the length direction of the first terminal fitting. The holding force between the first and second electrical contact portions can be set to be larger than the force necessary for the resilient deformation of the first stretchable portion, using the friction force.

(32) (15) Preferably, either one of the first and second electrical contact portions includes a fitting portion, the other of the first and second electrical contact portions includes a fit portion, the fitting portion is fit to the fit portion in a direction intersecting the length direction of the first terminal fitting, and the holding force between the first and second electrical contact portions includes an engaging force between the fitting portion and the fit portion engaged in the length direction of the first terminal fitting. The holding force between the first and second electrical contact portions can be set to be larger than the force necessary for the resilient deformation of the first stretchable portion, using the engaging force between the fitting portion and the fit portion engaged in the length direction of the first terminal fitting. The holding force between the first and second electrical contact portions can be stably set to be large as compared to the case where the engaging force between the fitting portion and the fit portion is not used.

Details of Embodiment of Present Disclosure

(33) Specific examples of a connector and a connector device of the present disclosure are described below with reference to the drawings. Note that the present disclosure is not limited to these illustrations, but is represented by claims and intended to include all changes in the scope of claims and in the meaning and scope of equivalents.

Embodiment

(34) Hereinafter, a connector device **10** of one embodiment of the present disclosure is described with reference to FIGS. **1** to **10**. As shown in FIGS. **1** and **2**, the connector device **10** is provided with a first connector **14** as a connector provided in a first in-vehicle device **12** and a second connector **18** provided in a second in-vehicle device **16**. The connector device **10** can be arranged in an arbitrary orientation, but an upward direction, a leftward direction and a forward direction are described below based on an upward direction, a leftward direction and a forward direction shown in figures. Further, for a plurality of identical members, only some members may be denoted by a reference sign and the other members may not be denoted by the reference sign.

(35) <First Terminal Fittings **20**>

(36) The first connector **14** is structured such that two first terminal fittings **20**, **20** are accommodated in a first connector housing **22**. That is, the first connector **14** is provided with the two first terminal fittings **20**, **20** and the first connector housing **22** for accommodating the first terminal fittings **20**, **20**. As also shown in FIG. **3**, the first terminal fitting **20** is a busbar made of a plate-like electrically conductive metal material. The first terminal fitting **20** is a resilient member allowed to be resiliently deflected and deformed in a thickness direction. The first terminal fitting **20** is long in a front-rear direction, a front end part serves as a first device-side connecting portion

24 and a rear end part serves as a first electrical contact portion 26. That is, the first terminal fitting 20 includes the first device-side connecting portion 24 and the first electrical contact portion 26 arranged on a side opposite to the first device-side connecting portion 24 in a length direction of the first terminal fitting 20. The first device-side connecting portion 24 is formed into an annular disk shape and includes a first bolt hole 28 penetrating in a vertical direction, which is the thickness direction. The first electrical contact portion 26 includes a dome-shaped contact point portion 30 projecting upward. The contact point portion 30 may be formed by making the first electrical contact portion 26 partially thicker, but is formed by press-working to be partially convex upward and the lower surface thereof is concavely dented. A rear end edge part of the first electrical contact portion 26 serves as a guiding portion 32 inclined downward toward the rear.

(37) A first stretchable portion 34 is provided between the first device-side connecting portion 24 and the first electrical contact portion 26 in the first terminal fitting 20. The first stretchable portion 34 is formed by bending the first terminal fitting 20 in a plate thickness direction. In other words, the first stretchable portion 34 is formed by bending the first terminal fitting 20 about axes along a width direction of the first terminal fitting 20. Note that the width direction of the first terminal fitting 20 is a lateral direction. Further, the width direction of the first terminal fitting 20 is a direction orthogonal to the length direction of the first terminal fitting 20. Further, the width direction of the first terminal fitting 20 is a direction orthogonal to the plate thickness direction of the first terminal fitting 20. More particularly, the first stretchable portion 34 is in the form of a groove open upward and having a U-shaped cross-section, and connected to the first device-side connecting portion 24 and the first electrical contact portion 26 on an opening end part of this groove. In short, the first terminal fitting 20 includes the first stretchable portion 34, a device-side part 36 located forward of the first stretchable portion 34 and having the first device-side connecting portion 24, and a contact-side part 34 located rearward of the first stretchable portion 34 and having the first electrical contact portion 26. In other words, the first terminal fitting 20 includes the first stretchable portion 34 in an electrical coupling portion, which is an intermediate region connecting the first device-side connecting portion 24 and the first electrical contact portion 26. A pair of positioning pieces 42, 42 project outward in the lateral direction between the first device-side connecting portion 24 and the first stretchable portion 34 in the device-side part 36.

(38) The first stretchable portion 34 of this embodiment includes a pair of vertical walls expanding orthogonally to the front-rear direction and a bottom wall connecting the lower ends of the vertical walls. Since these vertical walls and bottom wall are respectively in the form of flat plates, bent parts 40 are provided at four positions. However, the specific shape of the first stretchable portion 33 is not limited. For example, in a lateral view, the vertical walls and the bottom wall may be formed into a curved shape to be smoothly connected without forming any angle, an expanded shape or contracted shape in which a facing distance between the vertical walls changes in the vertical direction, a V shape free from the bottom wall or the like. Heights in the vertical direction of the pair of vertical walls may be different from each other. In that case, for example, in an initial shape of the first terminal fitting 20 not deformed by the action of an external force, the first device-side connecting portion 24 and the first electrical contact portion 26 are arranged at mutually different positions in the vertical direction. Note that the bending of the first terminal fitting 20 is not necessarily limited to the bending of the first terminal fitting at an angle. For example, the bending of the first terminal fitting 20 is realized also by smooth nonangular curving. Therefore, the first stretchable portion may be, for example, formed into a curved chevron shape or a wavy curved shape in which convex curves and concave curves are alternately connected.

(39) The first terminal fitting 20 is stretchable to change a distance in the front-rear direction, which is a separation distance between the first device-side connecting portion 24 and the first electrical contact portion 26, by the resilient deformation of the first stretchable portion 34. That is, the first terminal fitting 20 is formed by a plate material resiliently deformable in the thickness direction, whereby the angles of the bent parts 40 at the four positions in the first stretchable

portion **34** can be elastically changed. By angle changes of the bent parts **40**, the first stretchable portion **34** is stretched to change a width in the front-rear direction of an upper opening. Thus, the distance in the front-rear direction between the first device-side connecting portion **24** and the first electrical contact portion **26** connected to the upper end edge of the first stretchable portion **34** changes. That is, the first electrical contact portion **26** is relatively displaceable with respect to the first device-side connecting portion **24** by the resilient deformation of the first stretchable portion **34**.

(40) In the first terminal fitting **20**, relative movements of the first device-side connecting portion **24** and the first electrical contact portion **26** in the vertical direction, accompanying a relative angle change of the device-side part **36** and the contact-side part **38**, are also allowed by the deformation of the first stretchable portion **34**. Note that the relative movements of the first device-side connecting portion **24** and the first electrical contact portion **26** in the vertical direction can be also allowed by the resilient deformation of the device-side part **36** and the contact-side part **38** in the thickness direction.

(41) The first stretchable portion **34** is constituted by the pair of vertical walls and a horizontal wall in the form of flat plates, and the bent parts **40** at the four positions are respectively bent substantially at a right angle. Thus, as compared to the case where the entire first stretchable portion is smoothly curved, the first stretchable portion **34** is easily extended and contracted by angle changes between the vertical walls and the horizontal wall or an angle change between the vertical wall and the device-side part **36** or the contact-side part **38**.

(42) The first stretchable portion **34** allowing relative displacements of the first device-side connecting portion **24** and the first electrical contact portion **26** in the front-rear direction and vertical direction as described above can be easily provided by bending the first terminal fitting **20** constituted by a busbar, which is a single component, in the plate thickness direction. Moreover, since the first stretchable portion **34** is integrally provided by a part of the first terminal fitting **20**, an increase in the number of components is avoided and a cost reduction is expected.

(43) <Leaf Spring Clip **44**>

(44) A leaf sprig clip **44** serving as a resilient member is mounted on the first electrical contact portion **26** of the first terminal fitting **20**. The leaf sprig clip **44** includes a pair of flat plate portions **46**, **46** in the form of rectangular plates facing each other in the vertical direction with a gap therebetween and a coupling plate portion **48** coupling the pair of flat plate portions **46** to each other at one side edge parts (first side edge parts) in the lateral direction. The leaf sprig clip **44** is made of a resilient material such as a metal. In the leaf sprig clip **44**, a facing distance between the pair of flat plate portions **46**, **46** at a position distant from the first side edge parts is changeable by elastic angle changes between the flat plate portions **46**, **46** and the coupling plate portion **48**. The leaf sprig clip **44** may be formed of an electrically insulating material, but is preferably formed of an electrically conductive material.

(45) Each of the pair of flat plate portions **46**, **46** includes a sandwiching protrusion **50** projecting inward in a facing direction. The sandwiching protrusion **50** extends in the front-rear direction while having an arcuate cross-section. The sandwiching protrusion **50** can be, for example, formed by partially denting the flat plate portion **46** through press-working, similarly to the contact point portion **30** of the first terminal fitting **20**. The upper flat plate portion **46** is provided with a pair of guide pieces **52**, **52** projecting outward in the front-rear direction. The guide pieces **52**, **52** are inclined upward toward projecting tips from the flat plate portion **46**. The guide piece **52** may be provided only on a rear side of the flat plate portion **46** to obtain a function of guiding a second terminal fitting **92** to be described later. The guide pieces **52** may be provided on both front and rear sides of the flat plate portion **46**. If the guide pieces **52** are provided on both front and rear sides of the flat plate portion **46**, a pair of the left and right leaf sprig clips **44** can be commonly used.

(46) The leaf sprig clip **44** is mounted on the first electrical contact portion **26** of the first terminal

fitting **20** by inserting the first electrical contact portion **26** of the first terminal fitting **20** into between the pair of flat plate portions **46, 46** facing each other. The leaf sprig clip **44** may be fixed to the first terminal fitting **20**, for example, by fixing the lower flat plate portion **46** to the lower surface of the first electrical contact portion **26**. In this embodiment, the leaf spring clip **44** is not particularly positioned with respect to the first terminal fitting **20**. The leaf sprig clip **44** is prevented from being separated from the first terminal fitting **20** by being accommodated into the first connector housing **22** to be described later.

(47) <First Connector Housing **22**>

(48) The first connector housing **22** is in the form of a hollow box as a whole, and structured by combining a first member **54** and a second member **56** in the vertical direction. The first member **54** is in the form of a bottomed tube open upward as a whole and includes a peripheral wall **58** in the form of a rectangular tube and a partition wall portion **60** expanding orthogonally to the lateral direction in a lateral center. Since the first member **54** is bilaterally symmetrical with respect to the partition wall portion **60**, one of left and right structures with respect to the partition wall portion **60** is described here. A facing distance in the lateral direction between the peripheral wall **58** and the partition wall portion **60** of the first member **54** is larger than a width in the lateral direction of the first terminal fitting **20**.

(49) The first member **54** includes a first accommodation recess **62** for accommodating the first stretchable portion **34** of the first terminal fitting **20**. Positioning protrusions **64** projecting upward along the peripheral wall **58** from the bottom wall of the first member **54** are provided in both end parts in the lateral direction in a front end part of the first accommodation recess **62**. A facing distance in the lateral direction between the pair of positioning protrusions **64, 64** of the first accommodation recess **62** is substantially equal to a width in the lateral direction of the first stretchable portion **34**.

(50) As shown in FIG. 2, a terminal receiving portion **66** projecting forward at the upper end of the first accommodation recess **62** is provided forward of the first accommodation recess **62** in the first member **54**. As shown in FIG. 3, a front part of the peripheral wall **58** of the first member **54** includes projecting portions **67** projecting upward from the front end of the terminal receiving portion **66**. In other words, the terminal receiving portion **66** is provided in a stepwise manner in an intermediate part in the vertical direction of the front part of the peripheral wall **58** of the first member **54**. The projecting portions **67**, which are parts projecting upward from the terminal receiving portion **66** in the peripheral wall **58**, are facing the positioning protrusions **64** of the first accommodation recess **62** from front. A facing distance in the front-rear direction between the projecting portions **67** and the positioning protrusions **64** is substantially equal to a width direction in the front-rear direction of the positioning pieces **42** of the first terminal fitting **20**. The projecting portions **67** are partially provided at both ends and a center in the lateral direction of the first member **54**, and constitute device-side insertion portions **68** penetrating in the front-rear direction.

(51) A second accommodation recess **70** for accommodating the first electrical contact portion **26** of the first terminal fitting **20** is formed behind the first accommodation recess **62** in the first member **54**. The first accommodation recess **62** has a larger vertical depth than the second accommodation recess **70** to enable the accommodation of the first stretchable portion **34** projecting further downward than the first electrical contact portion **26**. A pair of first position defining portions **72, 72** projecting on the surfaces of the peripheral wall **58** and the partition wall portion **60** of the first member **54** facing each other in the lateral direction are provided on a boundary part between the first and second accommodation recesses **62, 70**. A distance in the lateral direction between the first position defining portions **72** and **72** is substantially equal to a lateral width of the contact-side part **38** in the first terminal fitting **20**. Further, a pair of first position defining portions **74, 74** projecting on the surfaces of the peripheral wall **58** and the partition wall portion **60** facing each other in the lateral direction in the second accommodation recess **70** are provided behind the first position defining portions **72**. A distance in the lateral

direction between the second position defining portions **74** and **74** is substantially equal to a distance in the lateral direction between the first position defining portions **72** and **72**.

(52) In the peripheral wall **58**, a pair of insertion windows **76**, **76** penetrating in the front-rear direction are formed in a part constituting rear wall portions of the second accommodation recesses **70**. The insertion window **76** has such an opening shape that a front end part of the second terminal fitting **92** to be described later is insertable through the insertion window **76**. The insertion window **76** has a hole cross-section larger in the vertical direction and lateral direction than a cross-section of the second terminal fitting **92** along a direction orthogonal to the front-rear direction. The pair of insertion windows **76**, **76** are located on laterally outer sides of the partition wall portion **60**. The second accommodation recess **70** is open to outside through the insertion window **76**.

(53) Locking claws **78** are formed on the lateral outer surfaces of the peripheral wall **58**. The locking claw **78** projects outward in the lateral direction from the peripheral wall **58**. The locking claw **78** has a right triangular cross-section with a lower side substantially orthogonal to the vertical direction. A projecting height of the locking claw **78** is reduced toward an upper side. In this embodiment, two locking claws **78** are disposed at positions separated in the front-rear direction. Those two locking claws **78** are respectively provided on each of both left and right sides of the first member **54**. The number and arrangement of the locking claws **78** are not particularly limited.

(54) As shown in FIG. 2, the second member **56** is in the form of a plate capable of covering an upper opening of the first member **54**, and includes a third position defining portion **77** projecting downward. As shown in FIG. 3, locking pieces **80** projecting downward are provided on both ends in the lateral direction of the second member **56**. The locking piece **80** is plate-like and allowed to be deflected and deformed in the lateral direction, which is a thickness direction. The locking piece **80** is formed with a claw insertion hole **82** corresponding to the locking claw **78** of the first member **54** and penetrating in the lateral direction. In this embodiment, two locking piece **80** are disposed at positions separated in the front-rear direction. Those two locking pieces **80** are respectively provided on each of both left and right sides of the second member **56**. The number and arrangement of the locking pieces **80** can be changed as appropriate to correspond to the locking claws **78**.

(55) By overlapping the second member **56** on the upper surface of the first member **54**, the upper opening of the first member **54** is closed by the second member **56** and the first connector housing **22** of a hollow structure provided with a space inside is configured. By locking the locking pieces **80** of the second member **56** to the locking claws **78** of the first member **54**, the first and second members **54**, **56** are positioned to each other.

(56) <Assembling of First Connector **14**>

(57) The pair of first terminal fittings **20**, **20** and the pair of leaf spring clips **44**, **44** are accommodated into the first connector housing **22**. That is, the first stretchable portions **34** of the first terminal fittings **20** are inserted into the first accommodation recesses **62** of the first member **54** from above and the contact-side parts **38** of the first terminal fitting **20** are inserted into the second accommodation recesses **70** of the first member **54** from above. The device-side parts **36** of the first terminal fittings **20** have rear end parts overlapped and supported on the upper surface of the terminal receiving portion **66** in the device-side insertion portions **68** of the first member **54**, and the first device-side connecting portions **24** are located forward of the first member **54** through the device-side insertion portions **68**. The leaf spring clips **44** are accommodated in the second accommodation recesses **70** of the first member **54** while being mounted on the first electrical contact portions **26** of the first terminal fittings **20**. Then, the second member **56** is fixed to the first member **54** to close the upper opening of the first member **54** accommodating the first terminal fittings **20** and the leaf spring clips **44**. In this way, the first connector **14** structured to accommodate the first terminal fittings **20** and the leaf spring clips **44** in the first connector housing **22** is configured.

(58) The first terminal fitting **20** is positioned in the lateral direction with respect to the first

member **54** by disposing the first stretchable portion **34** between the pair of left and right positioning protrusions **64** and **64**. Further, the first terminal fitting **20** is positioned in the lateral direction with respect to the first member **54** by inserting the device-side part **36** through the device-side insertion portion **68**. By inserting the positioning pieces **42** provided on the device-side part **36** into between the projecting portions **67** and the positioning protrusions **64** facing in the front-rear direction, the first terminal fitting **20** is positioned in the front-rear direction with respect to the first member **54**. Further, the first terminal fitting **20** is positioned in the lateral direction with respect to the first member **54** by disposing the contact-side part **38** between the pair of left and right first position defining portions **72**, **72**. Furthermore, the first terminal fitting **20** is positioned in the lateral direction with respect to the first member **54** by disposing the contact-side part **38** between the pair of left and right second position defining portions **74**, **74**.

(59) Note that, with the first terminal fitting **20** positioned with respect to the first connector housing **22** as described above, the first stretchable portion **34** of the first terminal fitting **20** accommodated in the first accommodation recess **62** is separated from the inner surface of the first accommodation recess **62** as shown in FIG. 2. In this way, the first stretchable portion **34** is allowed to be deformed and displaced in the first connector housing **22**. In this way, the extension and contraction of the first stretchable portion **34** are not limited by the first connector housing **22**. Further, the contact-side part **38** of the first terminal fitting **20** is positioned in the lateral direction with respect to the first connector housing **22** and allowed to be deformed and displaced in the vertical direction and front-rear direction. These allow the first electrical contact portion **26** accommodated in the first connector housing **22** to be displaced in the vertical direction and front-rear direction in the first connector housing **22**.

(60) The leaf spring clip **44** is not fixed with respect to the first connector housing **22** and is displaceable in the first connector housing **22** according to a displacement of the first electrical contact portion **26**. The leaf spring clip **44** is disposed behind the second position defining portions **74** and displacement amounts in the front-rear direction and lateral direction with respect to the first terminal fitting **20** are limited by the first member **54**. In this way, the detachment of the leaf spring clip **44** from the first terminal fitting **20** is prevented. A forward displacement amount of the leaf spring clip **44** with respect to the first connector housing **22** is limited also by the contact of the leaf spring clip **44** with the third position defining portion **77** provided in the second member **56**.

(61) <First In-Vehicle Device **12**>

(62) The first connector **14** structured as just described is provided in the first in-vehicle device **12** as shown in FIG. 2. The first in-vehicle device **12** is, for example, a motor of an automotive vehicle and structured to accommodate first connection terminals **86** serving as first terminal portions inside a first case **84**. The first case **84** is formed with a first window portion **88** penetrating through a rear wall portion in the front-rear direction. A front part of the first connector housing **22** is accommodated in the first case **84**. The first device-side connecting portions **24** of the first terminal fittings **20** projecting forward from the first connector housing **22** are overlapped on the first connection terminals **86**. The overlapped first device-side connecting portions **24** and first connection terminals **86** are fixed in an electrically connected state by first bolts **90** inserted through the first bolt holes **28** of the first device-side connecting portions **24**. In this way, the first connector **14** is provided in the first in-vehicle device **12** in a state where the first terminal fittings **20** are conductive with the first connection terminals **86** of the first in-vehicle device **12**. The first connector housing **22** projects rearward from the first case **84** through the first window portion **88**.

(63) <Second Connector **18** (Second Terminal Fittings **92**)>

(64) The second connector **18** is provided with the second terminal fittings **92**. As also shown in FIG. 3, the second terminal fitting **92** is a busbar plate-like and made of electrically conductive metal or the like, similarly to the first terminal fitting **20**. Note that the second terminal fitting **92** is thicker than the first terminal fitting **20**. In other words, the first terminal fitting **20** is thinner than the second terminal fitting **92**. Further, the second terminal fitting **92** is longer in the front-rear

direction than the first terminal fitting **20**. The second terminal fitting **92** is allowed to be resiliently deflected and deformed in the vertical direction, which is a thickness direction. In this way, the second terminal fitting **92** is allowed to be deformed in the thickness direction by a simple configuration by the busbar, which is a single component. A rear end part of the second terminal fitting **92** includes a second device-side connecting portion **94**, and a front end part thereof includes a second electrical contact portion **96** serving as a mating electrical contact portion. The second device-side connecting portion **94** is wider in the lateral direction than the other part of the second terminal fitting **92**, and includes a second bolt hole **98** penetrating in the vertical direction. A front end part of the second electrical contact portion **96** is formed into a tapered shape to be narrower in the lateral direction toward the front. Note that a front end part of the second electrical contact portion **96** may be thinner in the vertical direction toward the front.

(65) <Second In-Vehicle Device **16**>

(66) The second device-side connecting portion **94** of the second terminal fitting **92** constituting the second connector **18** is connected to the second in-vehicle device **16** as shown in FIG. **2**. The second in-vehicle device **16** is a device to be electrically connected to the first in-vehicle device **12** and, for example, an inverter or ECU (Electric Control Unit) of the automotive vehicle. The second in-vehicle device **16** is structured to accommodate second connection terminals **102** serving as second terminal portions inside a second case **100** serving as a casing. The second case **100** is formed with a second window portion **104** serving as an insertion hole penetrating through a front wall portion in the front-rear direction. Then, the second terminal fittings **92** are inserted through the second window portion **104** of the second case **100** of the second in-vehicle device **16**. The second device-side connecting portions **94** of the second terminal fittings **92** are overlapped on the second connection terminals **102**. The second device-side connecting portions **94** and the second connection terminals **102** are fixed in an electrically connected state by second bolts **106** inserted through the second bolt holes **98**. In this way, the second connector **18** is provided in the second in-vehicle device **16** in a state where the second terminal fittings **92** are conductive with the second connection terminals **102** of the second in-vehicle device **16**. The second electrical contact portions **96** project forward to outside from the second case **100** through the second window portion **104**. The second electrical contact portion **96** is made displaceable in the vertical direction by the resilient deflection and deformation of an intermediate region of the second terminal fitting **92** located between the second device-side connecting portion **94** and the second electrical contact portion **96**. In other words, the second electrical contact portion **96** is relatively displaceable with respect to the second device-side connecting portion **94** by the resilient deformation of the second terminal fitting **92**.

(67) <Connection of First Connector **14** and Second Connector **18**>

(68) A front end part of the second terminal fitting **92** extending out forward through the second window portion **104** from the second case **100** of the second in-vehicle device **16** is inserted into the insertion window **76** of the first connector housing **22**. The second electrical contact portion **96** of the second terminal fitting **92** inserted into the first connector housing **22** is inserted between the upper flat plate portion **46** of the leaf spring clip **44** and the first electrical contact portion **26** of the first terminal fitting **20** in the vertical direction as shown in FIGS. **2** and **4**. The first and second electrical contact portions **26**, **96** are held in a state pressed against each other by the resilience of the leaf spring clip **44**. In other words, the first and second electrical contact portions **26**, **96** are held in a contact state by the resilience of the leaf spring clip **44**. In particular, the first and second electrical contact portions **26**, **96** in the state contacting in the vertical direction are sandwiched in the vertical direction by the pair of flat plate portions **46**. More particularly, as shown in FIG. **5**, the first and second electrical contact portions **26**, **96** in the state contacting in the vertical direction are sandwiched in the vertical direction by the sandwiching protrusions **50** of the pair of flat plate portions **46**. In this way, the first and second terminal fittings **20**, **92** are conductively connected. In this way, the first and second in-vehicle devices **12**, **16** are electrically connected by the connector

device **10** composed of the first and second connectors **14**, **18**.

(69) Further, a holding force between the first and second electrical contact portions **26**, **96** held as described above is set to be larger than a force necessary for the resilient deformation of the first stretchable portion **34**. In other words, when the first and second in-vehicle devices **12**, **16** in a state assembled with each other are relatively displaced due to vibration or the like, the first stretchable portion **34** is set to be resiliently deformed earlier than relative displacements of the first and second electrical contact portions **26**, **96** in the front-rear direction. The holding force between the first and second electrical contact portions **26**, **96** includes friction forces between members contacting in a direction intersecting the length direction of the first terminal fitting **20**. In particular, the holding force between the first and second electrical contact portions **26**, **96** includes a friction force between the first and second electrical contact portions **26**, **96**, a friction force between the first electrical contact portion **26** and the lower flat plate portion **46** and a friction force between the second electrical contact portion **96** and the upper flat plate portion **46**. Note that, in FIG. **4**, the first and second terminal fittings **20**, **92** on one side are connected with the second member **56** of the first connector housing **22** removed from the first member **54**. Further, in FIG. **4**, the first and second terminal fittings **20**, **92** and the leaf spring clip **44** on the other side are shown in a disassembled state.

(70) The rear end surface of the leaf spring clip **44** is constituted by the guide piece **52** and the sandwiching protrusions **50**. The rear end surface of the leaf spring clip **44** is formed into an inclined surface extending in the vertical direction toward the rear. In this way, if the second terminal fitting **92** contacts the leaf spring clip **44** from behind, a component force of a contact reaction force acts upward on the leaf spring clip **44**, and the upper flat plate portion **46** is moved upward with respect to the first electrical contact portion **26** of the first terminal fitting **20**. As a result, a gap is formed between the upper flat plate portion **46** of the leaf spring clip **44** and the first electrical contact portion **26**, and the second electrical contact portion **96** of the second terminal fitting **92** can be inserted into this gap.

(71) If position shifts occur due to a dimensional error or the like between the first and second in-vehicle devices **12**, **16** in connecting the first in-vehicle device **12** provided with the first connector **14** and the second in-vehicle device **16** provided with the second connector **18**, these position shifts are allowed by the connector device **10**.

(72) That is, if the positions of the first and second in-vehicle devices **12**, **16** are shifted in the vertical direction, the first terminal fitting **20** in the first connector **14** is resiliently deformed at the first stretchable portion **34**, whereby the first electrical contact portion **26** relatively moves in the vertical direction with respect to the first device-side connecting portion **24** up to a position where the first and second electrical contact portions **26**, **96** can be connected. Note that the position shifts in the vertical direction are allowed also by the resilient deformation of at least one of the device-side part **36** and the contact-side part **38** of the first terminal fitting **20** in the thickness direction. Further, since the second terminal fitting **92** is a busbar, the position shifts of the first and second in-vehicle devices **12**, **16** in the vertical direction are allowed also by the resilient deformation of the second terminal fitting **92** in the thickness direction.

(73) If relative positions of the first and second in-vehicle devices **12**, **16** are shifted in the front-rear direction, the first terminal fitting **20** is resiliently deformed at the first stretchable portion **34**, whereby a distance in the front-rear direction between the first device-side connecting portion **24** and the first electrical contact portion **26** changes and position shifts of the first and second in-vehicle devices **12**, **16** in the front-rear direction are allowed. Further, since the second terminal fitting **92** extends in the front-rear direction and is inserted into the first connector **14** in the front-rear direction, the position shifts of the first and second in-vehicle devices **12**, **16** in the front-rear direction are allowed also by a change in the insertion amount of the second terminal fitting **92** into the first connector **14**.

(74) Further, since the second electrical contact portion **96** of the second terminal fitting **92** is in the

form of a flat plate, even if the positions of the first and second in-vehicle devices **12**, **16** are shifted in the lateral direction, the first and second electrical contact portions **26**, **96** can be connected over the width of the second electrical contact portion **96**.

(75) For example, as shown in FIG. **6**, the first and second electrical contact portions **26**, **96** are maintained in the connected state even if center positions thereof in the lateral direction are shifted.

(76) Further, if the first and second in-vehicle devices **12**, **16** are inclined in a circumferential direction about a center axis extending in the front-rear direction, the second electrical contact portion **96** can contact the contact point portion **30** and the first and second electrical contact portions **26**, **96** can be connected since the contact point portion **30** of the first electrical contact portion **26** has a dome shape. For example, even if the second electrical contact portion **96** is inclined about the center axis extending in the front-rear direction with respect to the first electrical contact portion **26** as shown in FIG. **7**, the connected state of the first electrical contact portion with the contact point portion **30** is maintained. Since the sandwiching protrusions **50** of the leaf spring clip **44** have an arcuate circular cross-section, the first or second electrical contact portion **26**, **96** and the leaf spring clip **44** are allowed to be inclined. Also by that, the first and second electrical contact portions **26**, **96** can be set in a contact state to be inclined about the center axis extending in the front-rear direction.

(77) As just described, according to the connector device **10**, since the first terminal fittings **20** of the first connector **14** are busbars, relative position shifts of the first and second in-vehicle devices **12**, **16** in a vehicle mounted state are allowed by the deformation or the like of the first terminal fittings **20**. Further, in this embodiment, since the second terminal fittings **92** of the second connector **18** are also busbars, the relative position shifts of the first and second in-vehicle devices **12**, **16** in the vehicle mounted state are allowed by the deformation or the like of the second terminal fittings **92**. Moreover, as described above, the relative position shifts of the first and second in-vehicle devices **12**, **16** are allowed with a large degree of freedom including twists and the like by the contact point portions **30**, the sandwiching protrusions **50** of the leaf spring clips **44** and the like. Therefore, the damage of the connector device **10**, the first in-vehicle device **12**, the second in-vehicle device **16** and the like due to the continuous action of stresses associated with the position shifts, a connection failure of the first and second connectors **14**, **18** and the like are prevented.

(78) Since the connector device **10** allows relative displacements of the first and second in-vehicle devices **12**, **16**, the relative displacements of the first and second in-vehicle devices **12**, **16** are allowed by the connector device **10** also when vibration due to the travel of the vehicle or the like is applied between the first and second in-vehicle devices **12**, **16**. Therefore, the damage of the connector device **10**, the first in-vehicle device **12**, the second in-vehicle device **16** and the like due to an input during travel is avoided. Particularly, in the structure of this embodiment provided with the first stretchable portions **34**, the relative displacements of the first and second in-vehicle devices **12**, **16** are more easily allowed by the deformation of the first stretchable portions **34**, and followability to a vibration input can be improved.

(79) The leaf spring clip **44** for holding the first electrical contact portion **26** of the first terminal fitting **20** and the second electrical contact portion **96** of the second terminal fitting **92** in the contact state is displaceable inside the first connector housing **22**. In this way, relative displacements of the first device-side connecting portion **24** and the first electrical contact portion **26** and those of the second device-side connecting portion **94** and the second electrical contact portion **96** are allowed without being hindered by the leaf spring clip **44**. Therefore, relative displacements associated with position shifts of the first and second in-vehicle devices **12**, **16** and a vibration input can be effectively allowed by the connector device **10**. Further, relative displacements of the first and second electrical contact portions **26**, **96** can be allowed while stably maintaining the contact state of the first and second electrical contact portions **26**, **96** by a simple structure of disposing the leaf spring clip **44** in the first connector housing **22** with a gap

therebetween.

(80) Next, effects of the above embodiment are complemented.

(81) The first electrical contact portion **26** is relatively displaceable with respect to the first device-side connecting portion **24**.

(82) For example, as shown in FIG. **8**, the angles of the bent parts **40** at the four positions of the first stretchable portion **34** can be changed by resilient deformation in the first terminal fitting **20**. In this way, the first electrical contact portion **26** is relatively displaceable in the front-rear direction with respect to the first device-side connecting portion **24**.

(83) Further, for example, as shown in FIGS. **9** and **10**, the angle of the bent part **40** at a boundary between the first stretchable portion **34** and the contact-side part **38** can be changed by resilient deformation in the first terminal fitting **20**. Note that FIGS. **9** and **10** schematically show a state where only the bent part **40** at the boundary between the first stretchable portion **34** and the contact-side part **38** is resiliently deformed. In this way, the first electrical contact portion **26** is relatively displaceable in the vertical direction with respect to the first device-side connecting portion **24**.

(84) Further, in the first terminal fitting **20**, each straight part can be deflected into a curved shape by being resiliently deformed when viewed from the lateral direction.

(85) The first electrical contact portion **26** is relatively displaceable in the front-rear direction and vertical direction with respect to the first device-side connecting portion **24** by combining the respective resilient deformations described above.

(86) Further, the second terminal fitting **92** can be deflected into a curved shape by resilient deformation when viewed from the lateral direction. Thus, the second electrical contact portion **96** is displaceable in the vertical direction with respect to the second device-side connecting portion **94**.

(87) Thus, even if the positions of the first and second in-vehicle devices **12**, **16** are shifted in the front-rear direction or shifted in the vertical direction or shifted in the vertical direction while being shifted in the front-rear direction, the first and second electrical contact portions **26**, **96** can be satisfactorily connected. Further, the first electrical contact portion **26**, the first stretchable portion **34** and the first device-side connecting portion **24** are constituted by a single component. Further, the second terminal fitting **92** is constituted by a single component. Note that the single component mentioned here is not a component obtained by joining a plurality of members. Thus, a reduction in the number of components and the simplification of a structure can be advantageously realized as compared to a conventional structure in which terminal fittings are constituted by joining a plurality of members including a braided wire and the like.

(88) Further, the holding force between the first and second electrical contact portions **26**, **96** is set to be larger than the force necessary for the resilient deformation of the first stretchable portion **34**. Thus, when the first and second in-vehicle devices **12**, **16** are relatively displaced due to vibration or the like caused by the travel of the vehicle, the first stretchable portion **34** is resiliently deformed rather than relative displacements of the first and second electrical contact portions **26**, **96** in the front-rear direction. Thus, the wear of the first and second electrical contact portions **26**, **96** is suppressed. Note that since the holding force between the first and second electrical contact portions **26**, **96** uses the friction forces between the members contacting in the direction intersecting the length direction of the first terminal fitting **20**, a special structure for generating a force other than the friction forces is unnecessary. Therefore, the first and second terminal fittings **20**, **92** need not be designed or worked into special structures.

(89) Since the first terminal fitting **20** is thinner than the second terminal fitting **92**, the first stretchable portion **34** is easily resiliently deformed. Further, the second terminal fitting **92** is hard to bend since being thicker than the first terminal fitting **20**. Particularly, since the second electrical contact portion **96** of this embodiment project to the outside of the second in-vehicle device **16**, the second electrical contact portion **96** may collide with another member before being connected to

the first electrical contact portion **26**. However, the bending of the second terminal fitting **92** is suppressed since the second terminal fitting **92** is thicker than the first terminal fitting **20**.

(90) <Modifications>

(91) Although the embodiment has been described in detail as a specific example of the present disclosure above, the present disclosure is not limited by this specific description. The present disclosure encompasses modifications, improvements and the like within such a range that the aim of the present disclosure can be achieved. For example, the following modifications of the embodiment are also included in the technical scope of the present disclosure.

(92) (1) The first terminal fitting **20** may be provided with a plurality of the first stretchable portions **34**. Further, the first stretchable portion **34** may be absent in the first terminal fitting **20**. A first stretchable portion can also be constituted by one vertical wall linking the device-side part **36** and the contact-side part **38**. In this case, a first terminal fitting is cranked (stepped), and the first device-side connecting portion **24** and the first electrical contact portion **26** may be shifted in the vertical direction in an initial shape of the first terminal fitting.

(93) (2) The second terminal fitting **92** may be provided with a second stretchable portion similar to the first stretchable portion **34** of the first terminal fitting **20**, and may be structured to be stretchable in a separating direction of the second device-side connecting portion **94** and the second electrical contact portion **96**. Note that, in the case of providing the second stretchable portion, various shapes can be adopted similarly to the first stretchable portion **34**. Further, the second terminal fitting **92** may be provided with a plurality of second stretchable portions.

(94) (3) The width and thickness of the first terminal fitting **20** may change halfway. For example, in the first terminal fitting **20**, the width of the first stretchable portion **34** can be made smaller than those of the device-side part **36** and the contact-side part **38** to reduce the deformation stiffness of the first stretchable portion **34**. Further, for example, the first terminal fitting **20** can be provided with partial projecting parts such as the positioning pieces **42** or partial recesses open in side surfaces, and the first terminal fitting **20** can also be positioned in the first connector housing **22** using the projecting parts and the recesses. Cross-sectional shape changes of the terminal fitting as described above are allowed not only in the first terminal fitting **20**, but also in the second terminal fitting **92**.

(95) (4) The resilient member is not limited to the leaf spring clip **44** described in the above embodiment. For example, upper and lower coil springs for respectively biasing the overlapped first and second electrical contact portions **26**, **96** toward each other in the vertical direction can be provided, and the first and second electrical contact portions **26**, **96** can be held in the contact state by the resilience of these coil springs. In this case, the resilient member is configured to include the upper and lower coil springs.

(96) (5) The resilient member is not essential. For example, if either one of the first and second electrical contact portions **26**, **96** is a male terminal, the other is a female terminal and the first and second electrical contact portions **26**, **96** are connected by fitting those male and female terminals, the resilient member as a separate component needs not be provided.

(97) (6) Although the second connector **18** is constituted by the two second terminal fittings **92**, **92** in the above embodiment, a second connector may be structured such that the second terminal fittings **92**, **92** are accommodated in a second connector housing having a hollow structure. By providing the second connector housing, the two second terminal fittings **92**, **92** are easily integrally handled as the second connector.

(98) (7) The first connector housing **22** of the first connector **14** may be at least partially accommodated in the first case **84** as in the above embodiment or may be entirely disposed outside the first case **84** of the first in-vehicle device **12**.

(99) (8) Although the holding force between the first and second electrical contact portions **26**, **96** uses the friction forces between the members contacting in the direction intersecting the length direction of the first terminal fitting **20** in the above embodiment, the holding force may include

forces other than the friction forces.

(100) For example, as shown in FIGS. **11** and **12**, the holding force between the first and second electrical contact portions **26**, **96** may be set by changing the shapes of the first and second electrical contact portions **26**, **96**. In this example, the first electrical contact portion **26** includes a fitting portion **110** and the second electrical contact portion **96** includes a fit portion **112**.

(101) The fitting portion **110** is a protrusion provided on the upper surface of the first electrical contact portion **26**. The fitting portion **110** is entirely formed on the first electrical contact portion **26** in the lateral direction, i.e. the width direction. The fitting portion **110** has an arcuate shape when viewed from the width direction of the first electrical contact portion **26**. The first electrical contact portion **26** is press-worked to be partially convex upward, thereby forming the fitting portion **110**. Thus, the first electrical contact portion **26** includes a recess **114** corresponding to the fitting portion **110** in a lower surface.

(102) The fit portion **112** is a recess provided in the lower surface of the second electrical contact portion **96**. The fit portion **112** is entirely formed on the second electrical contact portion **96** in the lateral direction, i.e. the width direction. The second electrical contact portion **96** is press-worked to be partially convex upward, thereby forming the fit portion **112**. Thus, the second electrical contact portion **96** includes a protrusion **116** corresponding to the fit portion **112** on an upper surface.

(103) The fitting portion **110** is fit upward, which is a direction intersecting the length direction of the first terminal fitting **20**, into the fit portion **112**. In this way, the holding force between the first and second electrical contact portions **26**, **96** includes an engaging force between the fitting portion **110** and the fit portion **112** engaged in the length direction of the first terminal fitting **20**. The holding force between the first and second electrical contact portions **26**, **96** including the engaging force between the fitting portion **110** and the fit portion **112** is set to be larger than the force necessary for the resilient deformation of the first stretchable portion **34**.

(104) Also in this way, when the first and second in-vehicle devices **12**, **16** are relatively displaced due to vibration or the like caused by the travel of the vehicle, the first stretchable portion **34** is resiliently deformed earlier than relative displacements of the first and second electrical contact portions **26**, **96** in the front-rear direction. Thus, the wear of the first and second electrical contact portions **26**, **96** is suppressed. Further, in this example, the holding force between the first and second electrical contact portions **26**, **96** can be set larger than the force necessary for the resilient deformation of the first stretchable portion **34**, using the engaging force between the fitting portion **110** and the fit portion **112** engaged in the length direction of the first terminal fitting **20**. Thus, for example, as compared to the above embodiment in which the engaging force of the fitting portion **110** and the fit portion **112** is not used, the holding force between the first and second electrical contact portions **26**, **96** can be stably set to be large.

(105) (6) Although the first and second cases **84**, **100** are configured to be arranged away from each other in the front-rear direction in the above embodiment, there is no limitation to this. The first and second cases **84**, **100** may be arranged in proximity in the front-rear direction.

(106) For example, as shown in FIG. **13**, a first case **120** may be configured to accommodate the entire first connector housing **22**.

(107) Further, for example, as shown in FIG. **14**, a second case **122** may be configured to be able to substantially entirely accommodate the second electrical contact portions **96** and accommodate a rear part of the first connector housing **22**.

(108) In this way, the exposure of the connector device **10** can be prevented. As shown in FIGS. **2**, **13** and **14**, the connector device **10** may be configured to electrically connect the first connection terminal **86** serving as the first terminal portion inside the first case **84** and the second connection terminal **102** serving as the second terminal portion inside the second case **100** not via flexible wires such as stranded wires therebetween, but via the first and second terminal fittings **20**, **92**. The connector device **10** of the embodiment may be configured to electrically connect the first and second in-vehicle devices **12**, **16** without via flexible wires such as stranded wires therebetween.

The first device-side connecting portion **24** may be the base end or fixed end of the first terminal fitting **20**, and the first electrical contact portion **26** may be the tip or free end of the first terminal fitting **20**. In the first stretchable portion **34** of the first terminal fitting **20**, the bent parts **40** may be, for example, configured to limit a resilient deforming direction of the first terminal fitting **20**. For example, bending directions and/or bent positions of the bent parts **40** may be so determined that distances between the free end and the fixed end of the first terminal fitting **20** in the length direction and the thickness direction of the first terminal fitting **20** easily change, a distance between the free end and the fixed end of the first terminal fitting **20** in the width direction of the first terminal fitting **20** hardly changes, and the free end of the first terminal fitting **20** is hardly twisted with respect to the fixed end of the first terminal fitting **20**. As shown in FIGS. **2** to **4**, the first stretchable portion **34** may be formed by bending or curving the first terminal fitting **20** at bending angles, which can be right angles, obtuse angles or acute angles, in the plate thickness direction at a plurality of positions between the both end parts (e.g. first device-side connecting portion **24** and the first electrical contact portion **26**) of the first terminal fitting **20**. As shown in FIG. **3**, the device-side part **36** located forward of the first stretchable portion **34** and including the first device-side connecting portion **24** and the contact-side part **38** located rearward of the first stretchable portion **34** and including the first electrical contact portion **26** in the first terminal fitting **20** may extend in parallel to each other. When the first terminal fitting **20** is in a natural state without being resiliently deformed as shown in FIG. **3**, the first device-side connecting portion **24** and the first electrical contact portion **26** may, for example, extend on a common virtual plane. The first device-side connecting portion **24** and the first electrical contact portion **26** may be located on two parallel virtual planes having different heights. As shown in FIGS. **5** to **7**, the pair of flat plate portions **46** of the leaf spring clip **44** can include the sandwiching protrusions **50** and non-projecting parts not formed with the sandwiching protrusions **50**, and the non-projecting parts of the flat plate portions **46** may face the first and second electrical contact portions **26**, **96** with gaps therebetween with the first and second electrical contact portions **26**, **96** sandwiched by the sandwiching protrusions **50**. These gaps are advantageous in maintaining the contact of the first and second electrical contact portions **26**, **96** with the sandwiching protrusions **50** while allowing the inclination of the first or second electrical contact portion **26**, **96** with respect to the leaf spring clip **44**. The first stretchable portion **34** of the embodiment may be referred to as a bent and deformed portion. The device-side part **36** and the contact-side part **38** may be referred to as a first straight portion and a second straight portion. The sandwiching protrusion **50** may be referred to as a ridge portion.

(109) Embodiments of the present disclosure are listed and described below.

(110) (1) A connector of the present disclosure is provided with a first terminal fitting including a first device-side connecting portion constituted by a busbar, provided on one end side and to be connected to a terminal portion of a first in-vehicle device and a first electrical contact portion provided on the other end side and to be connected to a mating electrical contact portion, and a first connector housing for accommodating the first terminal fitting, and the first electrical contact portion is displaceably accommodated in the first connector housing.

(111) According to the connector of the present disclosure, the first terminal fitting is constituted by the busbar, which is a single component. Further, a position shift from the mating electrical contact portion can be absorbed using a displacement of the first electrical contact portion by a simple structure of displaceably accommodating the first electrical contact portion in the first connector housing. Thus, a reduction in the number of components and the simplification of the structure can be advantageously achieved as compared to a conventional structure in which a terminal fitting needs to be constituted by joining a plurality of members including a braided wire and a supporting member for supporting an electrical contact portion and a mechanism for displaceably holding the supporting member in a connector housing are necessary.

(112) (2) Preferably, the first terminal fitting includes a first stretchable portion formed by bending

a part between the first device-side connecting portion and the first electrical contact portion in a plate thickness direction and stretchable in a separating direction of the first device-side connecting portion and the first electrical contact portion, and the first stretchable portion is stretchably accommodated in the first connector housing. Since the first terminal fitting includes the extendable and contractible first stretchable portion in an intermediate region in a length direction of the first terminal fitting, position shifts of the first electrical contact portion and the mating electrical contact portion can be more advantageously absorbed. Moreover, the first stretchable portion can be easily provided by bending the first terminal fitting constituted by the busbar in the plate thickness direction, and configured without increasing the number of components. Further, by stretchably accommodating the first stretchable portion in the first connector housing, it is prevented that the extension and contraction of the first stretchable portion are limited by the first connector housing.

(113) (3) Preferably, a resilient member is provided which holds the first electrical contact portion in contact with the mating electrical contact portion, and the resilient member is accommodated in the first connector housing in a state displaceable according to a displacement of the first electrical contact portion. By including the resilient member for holding the first electrical contact portion constituted by the busbar and in the form of a flat plate in contact with the mating electrical contact portion, conduction stability between the both electrical contact portions can be more reliably maintained. Further, since the resilient member is accommodated in the first connector housing displaceably according to the displacement of the first electrical contact portion, it is advantageously prevented by the resilient member that a position shift absorbing function is hindered by the displacement of the first electrical contact portion.

(114) (4) Preferably, in (3) described above, the resilient member is constituted by a leaf spring clip including a pair of flat plate portions arranged to face each other with a gap therebetween and a coupling plate portion coupling one side edge parts of the pair of flat plate portions, the leaf spring clip is mounted on the first electrical contact portion with the first electrical contact portion inserted between the pair of flat plate portions and one of the pair of flat plate portions overlapped on a bottom surface of the first electrical contact portion, and the mating electrical contact portion is inserted between the pair of flat plate portions to be overlapped on the first electrical contact portion. In this way, the resilient member can be provided using the leaf spring clip. Moreover, a structure of accommodating the resilient member in the first connector housing displaceably according to the displacement of the first electrical contact portion can be constructed only by mounting the leaf spring clip on the first electrical contact portion. Therefore, the structure of the connector can be more simplified.

(115) (5) Preferably, the first electrical contact portion includes a dome-shaped contact point portion projecting toward the mating electrical contact portion. Since the contact point portion provided at the first electrical contact portion has a dome shape, a contact state between the both electrical contact portions can be stably maintained even if the mating electrical contact portion to be overlapped on the first electrical contact portion is twisted and displaced such as by being rolled.

(116) (6) A connector device of the present disclosure is provided with a first connector provided in a first in-vehicle device and a second connector provided in a second in-vehicle device and to be connected to the first connector, and the first connector is the connector of any one of (1) to (5) described above.

(117) According to the connector device of the present disclosure, since the first connector is the connector of any one of (1) to (5) described above, the both first and second connectors can be reliably connected by absorbing position shifts thereof, and the aforementioned functions and effects by the connector of the present disclosure can be similarly enjoyed.

(118) (7) Preferably, in (6) described above, the second connector includes a second terminal constituted by a busbar, the second terminal fitting includes a second device-side connecting portion provided on one end side and to be connected to a terminal portion of the second in-vehicle

device and a second electrical contact portion provided on the other end side and to be connected to the first electrical contact portion, and the second electrical contact portion projects outward in a displaceable state from an insertion hole provided in a casing of the second in-vehicle device. Since the second terminal fitting of the second connector is also constituted by the busbar and the second electrical contact portion projects outward in the displaceable state from the second in-vehicle device, position shifts between the respective electrical contact portions of the first and second connectors can be more advantageously absorbed and connection between the both electrical contact portions can be stably realized, using displacements of the both first and second electrical contact portions at the time of connecting the first and second connectors. Moreover, since the second terminal fitting is also constituted by the busbar, which is a single component, a reduction in the number of components and the simplification of the structure can be more advantageously achieved.

LIST OF REFERENCE NUMERALS

(119) **10** connector device **12** first in-vehicle device **14** first connector (connector) **16** second in-vehicle device **18** second connector **20** first terminal fitting **22** first connector housing **24** first device-side connecting portion **26** first electrical contact portion **28** first bolt hole **30** contact point portion **32** guiding portion **34** first stretchable portion **36** device-side part **38** contact-side part **40** bent part **42** positioning piece **44** leaf spring clip (resilient member) **46** flat plate portion **48** coupling plate portion **50** sandwiching protrusion **52** guide piece **54** first member **56** second member **58** peripheral wall **60** partition wall portion **62** first accommodation recess **64** positioning protrusion **66** terminal receiving portion **67** projecting portion **68** device-side insertion portion **70** second accommodation recess **72** first position defining portion **74** second position defining portion **76** insertion window **77** third position defining portion **78** locking claw **80** locking piece **82** claw insertion hole **84** first case **86** first connection terminal (first terminal portion) **88** first window portion **90** first bolt **92** second terminal fitting **94** second device-side connecting portion **96** second electrical contact portion (mating electrical contact portion) **98** second bolt hole **100** second case (casing) **102** second connection terminal (second terminal portion) **104** second window portion (insertion hole) **106** second bolt **110** fitting portion **112** fit portion **114** recess **116** protrusion **120** first case **122** second case

Claims

1. A connector capable of electrically connecting a first terminal portion of a first in-vehicle device and a flat plate-like mating electrical contact portion of a second in-vehicle device, comprising: a first terminal fitting; and a first connector housing for accommodating the first terminal fitting, the first terminal fitting including a first device-side connecting portion projecting with a projection length from the first connector housing to be connectable to the first terminal portion and a first electrical contact portion connectable to the mating electrical contact portion and arranged on a side opposite to the first device-side connecting portion in a length direction of the first terminal fitting, the first electrical contact portion and the first device-side connecting portion being constituted by a single component, the first electrical contact portion being relatively displaceable with respect to the first device-side connecting portion, the projection length of the first device-side connecting portion from the first connector housing not changing and the first electrical contact portion of the first terminal fitting being displaced in the first connector housing when the first electrical contact portion is relatively displaced with respect to the first device-side connecting portion, the connector comprising a resilient member to be accommodated into the first connector housing, the resilient member being capable of holding the first electrical contact portion in contact with the mating electrical contact portion and displaceable according to a displacement of the first electrical contact portion in the first connector housing, the resilient member including a pair of flat plate portions arranged to face each other with a gap therebetween and a coupling plate portion coupling the pair

of flat plate portions, the resilient member being capable of sandwiching the first electrical contact portion and the mating electrical contact portion in a contact state in mutually contacting directions by the pair of flat plate portions, each of the pair of flat plate portions including a sandwiching protrusion projecting inward in a facing direction, the sandwiching protrusions being capable of sandwiching the first electrical contact portion and the mating electrical contact portion in the contact state in the mutually contacting directions, the first electrical contact portion including a contact point portion connectable to the mating electrical contact portion, the contact point portion being a circular dome-shaped projection having a diameter in a plan view, each of the sandwiching protrusions of the pair of flat plate portions of the resilient member being an elongated protrusion having an arcuate cross-section, one of the sandwiching protrusions of the pair of flat plate portions being configured to contact the flat plate-like mating electrical contact portion with a contact length in the length direction of the first terminal fitting with the resilient member sandwiching the first electrical contact portion and the mating electrical contact portion, and the sandwiching protrusion having the arcuate cross-section of the resilient member and the circular dome-shaped contact point portion of the first electrical contact portion allowing the flat plate-like mating electrical contact portion to be twisted and displaced in an out-of-plane direction of the flat plate-like mating electrical contact portion with the resilient member sandwiching the first electrical contact portion and the mating electrical contact portion.

2. The connector of claim 1, wherein: the first terminal fitting includes a first stretchable portion between the first device-side connecting portion and the first electrical contact portion, and the first electrical contact portion is relatively displaceable with respect to the first device-side connecting portion by resilient deformation of the first stretchable portion.

3. The connector of claim 2, wherein the first stretchable portion is formed by bending the first terminal fitting.

4. A connector device for electrically connecting a first terminal portion of a first in-vehicle device and a second terminal portion of a second in-vehicle device, comprising: a first connector mountable into the first in-vehicle device; and a second connector mountable into the second in-vehicle device, the first connector including a first terminal fitting and a first connector housing for accommodating the first terminal fitting, the second connector including a second terminal fitting, the second terminal fitting including a second device-side connecting portion connectable to the second terminal portion and a flat plate-like second electrical contact portion to be connected to the first terminal fitting and arranged on a side opposite to the second device-side connecting portion, the first terminal fitting including a first device-side connecting portion projecting with a projection length from the first connector housing to be connectable to the first terminal portion and a first electrical contact portion to be connected to the second electrical contact portion and arranged on a side opposite to the first device-side connecting portion in a length direction of the first terminal fitting, the first electrical contact portion and the first device-side connecting portion being constituted by a single component, the first electrical contact portion being relatively displaceable with respect to the first device-side connecting portion, the projection length of the first device-side connecting portion from the first connector housing not changing and the first electrical contact portion of the first terminal fitting being displaced in the first connector housing when the first electrical contact portion is relatively displaced with respect to the first device-side connecting portion, the first connector including a resilient member to be accommodated into the first connector housing, the resilient member being capable of holding the first electrical contact portion in contact with the second electrical contact portion and displaceable according to a displacement of the first electrical contact portion in the first connector housing, the resilient member including a pair of flat plate portions arranged to face each other with a gap therebetween and a coupling plate portion coupling the pair of flat plate portions, the resilient member being capable of sandwiching the first electrical contact portion and the second electrical contact portion in a contact state in mutually contacting directions by the pair of flat plate portions, each of the pair of flat plate

portions including a sandwiching protrusion projecting inward in a facing direction, the sandwiching protrusions being capable of sandwiching the first electrical contact portion and the second electrical contact portion in the contact state in the mutually contacting directions, the first electrical contact portion including a contact point portion connectable to the second electrical contact portion, the contact point portion being a circular dome-shaped projection having a diameter in a plan view, each of the sandwiching protrusions of the pair of flat plate portions of the resilient member being an elongated protrusion having an arcuate cross-section, one of the sandwiching protrusions of the pair of flat plate portions being configured to contact the flat plate-like second electrical contact portion with a contact length in the length direction of the first terminal fitting with the resilient member sandwiching the first electrical contact portion and the second electrical contact portion, and the sandwiching protrusion having the arcuate cross-section of the resilient member and the circular dome-shaped contact point portion of the first electrical contact portion allowing the flat plate-like second electrical contact portion to be twisted and displaced in an out-of-plane direction of the flat plate-like second electrical contact portion with the resilient member sandwiching the first electrical contact portion and the second electrical contact portion.

5. The connector device of claim 4, wherein: the second electrical contact portion and the second device-side connecting portion are constituted by a single component, and the second electrical contact portion is relatively displaceable with respect to the second device-side connecting portion.

6. The connector device of claim 5, wherein the second electrical contact portion projects to outside of the second in-vehicle device with the second connector mounted in the second in-vehicle device.

7. The connector device of claim 4, wherein: the first terminal fitting includes a first stretchable portion between the first device-side connecting portion and the first electrical contact portion, the first electrical contact portion is relatively displaceable with respect to the first device-side connecting portion by resilient deformation of the first stretchable portion, and the first stretchable portion is formed by bending the first terminal fitting.

8. The connector device of claim 7, wherein the first terminal fitting is thinner than the second terminal fitting.

9. The connector device of claim 7, wherein a holding force between the first and second electrical contact portions is set to be larger than a force necessary for the resilient deformation of the first stretchable portion.

10. The connector device of claim 9, wherein the holding force between the first and second electrical contact portions includes a friction force between members contacting in a direction intersecting the length direction of the first terminal fitting.

11. The connector device of claim 9, wherein: either one of the first and second electrical contact portions includes a fitting portion, the other of the first and second electrical contact portions includes a fit portion, the fitting portion is fit to the fit portion in a direction intersecting the length direction of the first terminal fitting, and the holding force between the first and second electrical contact portions includes an engaging force between the fitting portion and the fit portion engaged in the length direction of the first terminal fitting.
